

Numerical Analysis PhD Exam, September 2005

Do any eight of the ten problems below.

1. For solving $dy/dt = f(t, y)$, consider a one-step method of the form

$$y_1 = y_0 + h\Phi(t_0, y_0, h).$$

For any t_0 , it computes an approximation y_1 to $y(t_0 + h)$, given $y(t_0) = y_0$. Assuming enough smoothness on f , study the local truncation error of Runge–Kutta methods for which

$$\Phi(t_0, y_0, h) = \alpha_1 k_1 + \alpha_2 k_2,$$

where

$$k_1 = f(t_0, y_0),$$

and

$$k_2 = f(t_0 + \mu h, y_0 + \mu h k_1).$$

Under what conditions on the constants α_1, α_2 , and μ do we get second order methods?

2. Prove that any polynomial q of degree at most $n - 1$ satisfies

$$\sum_{i=0}^n q(x_i) \prod_{\substack{j=0 \\ j \neq i}}^n \frac{1}{x_i - x_j} = 0,$$

for any distinct $n + 1$ numbers x_i , $i = 0, 1, \dots, n$. (Suggestion: Identify the left hand side as a divided difference.)

3. Let $p_2(x)$ be the quadratic polynomial that interpolates $f(x)$ at nodes $x = 0, h$, and $2h$.

(a) Approximating

$$Q = \int_0^{3h} f(x) dx$$

by

$$Q_h = \int_0^{3h} p_2(x) dx,$$

derive a quadrature rule: Explicitly calculate w_1, w_2, w_3 so that

$$Q_h = w_1 f(0) + w_2 f(h) + w_3 f(2h).$$

(b) Assuming that $f(x)$ is four times continuously differentiable, show that

$$Q - Q_h = \frac{3}{8} h^4 f'''(0) + O(h^5).$$

4. Denote the complete (or clamped) and natural cubic splines interpolating $f \in C^2[a, b]$ at knots $t_0 < t_1 < \dots < t_n$ by $s_c(t)$ and $s_n(t)$, respectively.

- (a) Prove that

$$\int_a^b |s_c''(x)|^2 dx \leq \int_a^b |f''(x)|^2 dx.$$

State an analogous inequality for s_n .

- (b) Which of the two interpolating splines has smaller “energy”, i.e., which of $\int_a^b |s_c''(x)|^2 dx$ and $\int_a^b |s_n''(x)|^2 dx$ is smaller?

5. Let $p > 0$ and

$$x = \sqrt{p + \sqrt{p + \sqrt{p + \cdots}}},$$

where all the square roots are positive. Let $F(y) = \sqrt{p + y}$. (How is the fixed point of F related to x ?)

- (a) Consider the fixed point iteration $x_{n+1} = F(x_n)$. Prove that if the initial iterate x_0 satisfies $x_0 + p > 0$ then all iterates remain on the same side of x as x_0 .
 (b) Prove that the fixed point iteration converges for all choices of initial guesses greater than $-p+1/4$.
 (c) Whenever the fixed point iteration converges, what is its order of convergence?

6. This problem has three parts.

- (a) Consider the matrix $A = uv^*$ where u and $v \in \mathbb{C}^n$. Under what condition on u and v is A a projector?
 (b) Show that the Householder matrix $H = I - 2ww^*$ where $\|w\| = 1$ is unitary.
 (c) Given a vector $x \in \mathbb{C}^n$ and an integer k with $1 < k < n$, derive a formula for a Householder matrix with the property that $(Hx)_i = 0$ for $i > k$ and $(Hx)_i = x_i$ for $i < k$. Be sure to choose the signs so that the formula is numerically stable.

7. Consider the conjugate gradient algorithm for solving $Ax = b$ for a symmetric positive definite $A \in \mathbb{R}^{m \times m}$.

- (a) State a relationship between the n -th iterate of the algorithm and the best approximation to x from a Krylov space (be sure to specify the norm).
 (b) Prove that if A has only $n \leq m$ distinct eigenvalues then the iteration (in the absence of round off errors) converges to the exact solution x in at most n steps no matter what the initial iterate is. (You may use 7a.)

8. Let P and Q be two $m \times m$ orthogonal projectors.

- (a) Prove that $\|P - Q\|_2 \leq 1$.
 (b) Prove that if $\|P - Q\|_2 < 1$ then the ranges of P and Q have equal dimensions.

9. Let P and Q be Hermitian positive definite matrices. Prove that

$$x^*Px \leq x^*Qx, \quad \text{for all } x \in \mathbb{C}^n,$$

if and only if

$$x^*Q^{-1}x \leq x^*P^{-1}x, \quad \text{for all } x \in \mathbb{C}^n.$$

10. State the Rayleigh quotient iteration. Apply it to the 2×2 real matrix

$$A = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix},$$

where $\lambda_1 \neq \lambda_2$. Find the subset $S \subseteq \mathbb{R}^2$ having the property that the iteration applied to this matrix with initial guesses in S do not converge. Is S a set of measure zero?